A Note on 'On a problem of Brown, Erdős and Sós'

Sifei Li

June 25, 2026

1 Introduction

1.1 Problem background

We study the maximum number of edges in an r-uniform hypergraph on n vertices with no k edges spanning too few vertices.

Let $f^{(r)}(n; s, k)$ denote the maximum number of edges in an r-uniform hypergraph on n vertices that does not contain k edges spanning at most s vertices. For integers $r > t \geq 2$ and $k \geq 2$, when $s = k(r-t)+t$, the extremal function has order n^t [1]. Several works have therefore studied the normalized quantity

$$\pi(r, t, k) := lim_{n\to\infty} n^{-t} f^{(r)}(n; k(r-t)+t, k).$$

1.2 Known results and what this note does

In [3], Letzter and Sgueglia proved that, when k is even and r is sufficiently large, $\pi(r, t, k) = \frac{1}{t!}\binom{r}{t}^{-1}$. For odd k, they proved the existence of the limit in the cases $k = 5$ and $k = 7$. In the conclusion, they also sketched an argument showing that, for general odd k and sufficiently large r, the upper limit is at most

$$\frac{1}{t!} \cdot \frac{2}{2\binom{r}{t} - 1}.$$

This upper bound may be optimal: it agrees with the known limit for $k = 3$, proved in [2].

The purpose of this note is to provide a complete proof of the argument sketched in the conclusion of [3]. Specifically, our main result is the following:

Theorem 1. *Let k, t, r be positive integers, k be odd, $r-1 \geq t \geq 2$, there exists a constant $c_0(k, t)$ depending on k and t such that for all $r \geq c_0(k, t)$,*

$$\limsup_{n\to\infty} n^{-t} f^{(r)}(n; k(r-t)+t, k) \leq \frac{1}{t!} \frac{2}{2\binom{r}{t} - 1}.$$

We may assume $k \geq 5$, as the cases $k = 1$ is trivial and $k = 3$ is settled in [2].

1.3 Main technical content of the proof

The main technical part is the counting argument after decomposing the cleaned hypergraph into t-tight components. Components of size 1 and 2 are relatively direct. The main work is the treatment of components of size at least 3, where one has to count the t-sets contributed by each 2-configuration and then control the t-sets that may already be accounted for by edges or 2-configurations outside the component.

An earlier arXiv version of Letzter and Sgueglia contained a fuller proof of this part, but the counting for components of size at least 3 requires additional care. The proof given here follows the same general strategy, while replacing that final step with a corrected counting argument for large t-tight components.

This note should therefore be read as a detailed reconstruction of an omitted proof, rather than as a new asymptotic result beyond the statement indicated in Letzter–Sgueglia. A natural next step is to study the corresponding lower-bound problem, with the aim of narrowing the known range of $\pi(r,t,k)$ and ultimately determining its exact value.

2 Start of the note

2.1 Preliminaries

Notations. In an r-graph $\mathcal{F}$, we use $\mathcal{F}$ to represent the edge set $E(\mathcal{F})$, $V(\mathcal{F})$ to represent the vertex set of $\mathcal{F}$. $|\mathcal{F}|$ to represent the number of edges in $\mathcal{F}$, and $e \in \mathcal{F}$ to represent that e is an edge of $\mathcal{F}$.

For a positive integer s and k, define (s,k)-configuration to be a subgraph with k hyperedges and at most s vertices. We further shorten the notation, let k-configuration denote a $(k(r-t)+t,k)$-configuration, a k^--configuration denotes a $(k(r-t)+t-1,k)$-configuration. A hypergraph is k-free (k^--free) if it does not contain any k-configuration (k^--configuration).

A t-tight component is a collection of edges that can be ordered as $\{e_1,...,e_m\}$ such that for all i with $1<i\le m$, there exists an index $j<i$ such that $|V(e_j)\cap V(e_i)|\ge t$. We may use a component and a t-tight component interchangeably.

We remark some useful results that will be used in the proof of Theorem 1.

Lemma 1 (Lemma 2.6 in [3]). *Let r,k and t be fixed positive integers. Let $\mathcal{F}$ be a k-free n-vertex r-graph. Then there exists a subhypergraph $\mathcal{F}'$ of $\mathcal{F}$ such that the following holds.*

(P1) $\mathcal{F}'$ is ℓ^--free for every $\ell\in[2,k]$ with $\ell \mid (k-1)$ or $\ell \mid k$;

(P2) there is no 3^--configuration in $\mathcal{F}'$ which contains a 2 -configuration;

(P3) for every positive integers a and b with $a+b=k$, every a^--configuration and every b-configuration of $\mathcal{F}'$ are edge-disjoint;

(P4) $|\mathcal{F}'|\ge|\mathcal{F}|-O\left(n^{t-1}\right)$.

Claim 1 (Claim 3.2 in [3]). *Let $\mathcal{F}$ be a k-free r-graph and $e\in\mathcal{F}$. Then the number of 2-configurations of $\mathcal{F}$ containing e is at most $k-2$.*

Claim 2. *Suppose that r,t,k are integers that satisfy $t\ge2$, $k\ge5$, if r is large enough such that the following inequality holds,*

$$\left\{[2r-t(2k-3)]^t-2[r-(k-2)t]^t-\frac{3}{2}r^t\right\}\ge t!\left[(k-3)+(k-5)(k-2)\binom{4t-4}{t}-\frac{3}{4}\right]. \tag{1}$$

Denote $q=r-(k-2)t, \alpha=\frac{1}{2}\left(2\binom{r}{t}-1\right)$. Then

$$\binom{2q}{t}-2\binom{q}{t}-(k-3)-(k-5)(k-2)\binom{4t-4}{t}\ge\frac{3}{2}\alpha$$

Proof. Since $\frac{n^t}{t!}\ge\binom{n}{t}\ge\frac{(n-t)^t}{t!}$, we have

$$\begin{aligned}\binom{2q}{t}-2\binom{q}{t}-\frac{3}{2}\binom{r}{t}&\ge\frac{1}{t!}\left\{[2r-t(2k-3)]^t-2[r-(k-2)t]^t-\frac{3}{2}r^t\right\}\\&\ge(k-3)+(k-5)(k-2)\binom{4t-4}{t}-\frac{3}{4}\end{aligned}$$

given the assumption of r. Rearranging the equation gives the desired result. □

We note that the assumption of r is feasible when r is large enough, as when $t \geq 2$, the size of the left side is $Ar^t + O(r^{t-1})$ with some positive constant A, and the right side is just a constant with respect to r. Thus, we can simply choose the constant $c_0(k,t)$ in Theorem 1 to be any number such that inequality 1 holds.

2.2 Complete proof of the upper bound

In this section, we provide a complete proof of Theorem 1.

Proof of Theorem 1. Let F be a k-free n-vertex r-graph. By Lemma 1, $\exists \mathcal{F}' \subseteq \mathcal{F}$ such that $\mathcal{F}'$ satisfies $(P1), (P2), (P3)$ and $|\mathcal{F}'| \geq |\mathcal{F}| - O\left(n^{t-1}\right)$.

Let $\mathcal{F}_i$ be the set of edges lying in components of size i for $i = 1, 2$, and let $\mathcal{F}_3 := \mathcal{F}' \backslash (\mathcal{F}_1 \cup \mathcal{F}_2)$.

Denote $\mathcal{G}_i$ as the t-shadow of $\mathcal{F}_i$, for $i = 1, 2$. Denote $\mathcal{G}_3$ as the collection of t-sets T such that $T \notin \mathcal{G}_1 \cup \mathcal{G}_2$, and there is a unique component in $\mathcal{F}_3$ that contains a 2-configuration whose vertex set contains T.

Notice that $\mathcal{G}_1, \mathcal{G}_2, \mathcal{G}_3$ are disjoint. Let $\alpha = \frac{1}{2}\left(2\binom{r}{t} - 1\right)$.

We assert that for $i = 1, 2, 3, |G_i| \geq \alpha |F_i|$. If this is true, then

$$\begin{aligned}
|\mathcal{F}'| &= |\mathcal{F}_1| + |\mathcal{F}_2| + |\mathcal{F}_3| \\
&\leq \frac{1}{\alpha}\left(|\mathcal{G}_1| + |\mathcal{G}_2| + |\mathcal{G}_3|\right) \\
&\leq \frac{1}{\alpha}\binom{n}{t} \\
&= \frac{2\binom{n}{t}}{2\binom{r}{t} - 1} \\
&\leq \frac{2n^t}{t!\left(2\binom{r}{t} - 1\right)}
\end{aligned}$$

Therefore, $\limsup_{n\to\infty} n^{-t}|\mathcal{F}| \leq \limsup_{n\to\infty} n^{-t}\left(|\mathcal{F}'| + O\left(n^{t-1}\right)\right) \leq \frac{2}{t!\left(2\binom{r}{t}-1\right)}$.

The rest of the proof is for the assertion:

For $i = 1$, by definition $|\mathcal{G}_1| = \binom{r}{t}|\mathcal{F}_1| \geq \left(\binom{r}{t} - \frac{1}{2}\right)|\mathcal{F}_1| = \alpha |\mathcal{F}_1|$, for any $r \geq t$.

For $i = 2$, for every 2-configuration (e, f) where $e, f \in \mathcal{F}_2$, we have $|V(e) \cap V(f)| \geq t$, However, since $\mathcal{F}_2$ inherits the 2^--free property of $\mathcal{F}'$ obtained by $(P1)$ of Lemma 1, within $2r - t - 1$ vertices one can have at most one edge, which forces $|V(e) \cap V(f)| = t$. We will reuse this property a few times in the later proof. Thus, the number of choices of t-shadow of a 2-configuration (e, f) is exactly $2\binom{r}{t} - 1$. Multiplying by the number of 2-configurations in $\mathcal{F}_2$ results in

$$|\mathcal{G}_2| = \left(2\binom{r}{t} - 1\right)\frac{|\mathcal{F}_2|}{2} = \alpha |\mathcal{F}_2|.$$

For $i = 3$, note that by definition $\mathcal{F}_3$ can be partitioned into t-tight components, each of size at least 3. Let C be a component of $\mathcal{F}_3$. It suffices to show that $|\mathcal{G}_3(C)| \geq \alpha |C|$, where $\mathcal{G}_3(C)$ denotes the restriction of $\mathcal{G}_3$ to the t-sets that are contained in the vertex sets of the edges in C. If this holds for every component, then summing over all components gives the desired result.

By the definition of a t-tight component, there are at least $|C| - 1$ distinct 2-configurations contained in C.

We first state a few observations about C that will be used later:

- C is k-free and 2^--free, and no 3^--configuration in C contains a 2-configuration. These properties are inherited from $\mathcal{F}$ or follow from Lemma 1.
- Each 2-configuration in C spans exactly $2r - t$ vertices, by the 2^--free property.
- $|C| \le k - 1$, since otherwise this would violate the k-free property.

We partition $\mathcal{G}_3(C)$ into two disjoint collections of t-sets. The first collection $\mathcal{A}$ consists of all t-sets in $\mathcal{G}_3(C)$ that are fully contained in a single edge of a 2-configuration of C, while the second collection $\mathcal{B}$ consists of all t-sets T in $\mathcal{G}_3(C)$ for which there exists a 2-configuration (e_1, e_2) in C such that $T \cap \big(V(e_1) \setminus V(e_2)\big) \neq \emptyset$ and $T \cap \big(V(e_2) \setminus V(e_1)\big) \neq \emptyset$.

We only estimate the size of $\mathcal{B}$ and show that this is enough. For each edge e in C, it shares at most t vertices with any other edge in C, since otherwise the two edges would form a 2^--configuration. Using $|C| \le k - 1$, it follows that there are at least $q := r - t(k-2)$ vertices of e that are not shared with any other edge in C. We pick exactly q of them to form a subset Y_e of the vertex set.

For each 2-configuration $S = (e_1, e_2)$ in C, define $\mathcal{B}_S := \{T \subseteq Y_{e_1} \cup Y_{e_2} : |T| = t, T \cap Y_{e_1} \neq \emptyset, T \cap Y_{e_2} \neq \emptyset\}$. We have

$$|\mathcal{B}_S| = \binom{2q}{t} - 2\binom{q}{t},$$

where the term $2\binom{q}{t}$ subtracts the t-sets that are fully contained in Y_{e_1} or Y_{e_2}. It is also worth mentioning that for two distinct 2-configurations S and S', $\mathcal{B}_S \cap \mathcal{B}_{S'} = \emptyset$.

For any $T \in \mathcal{B}_S$, if T is not contained in any edge or 2-configuration outside C, then $T \in \mathcal{G}_3(C)$. Thus we still need to remove the t-sets in $\mathcal{B}_S$ that are contained in other components. These will be considered in the following assertions:

1. The number of t-sets T in (e_1, e_2) that are contained in an edge $e' \notin C$ is at most $(k-3)$.
2. The number of t-sets T in (e_1, e_2) that are contained in the vertex set of a 2-configuration (e', f') with $(e', f') \not\subseteq C$ is at most $(k-2)(k-5)\binom{4t-4}{t}$.

To prove assertion 1, suppose there is an edge $e' \notin C$ such that e' shares at least t vertices with (e_1, e_2). Then it must share exactly t vertices with (e_1, e_2), otherwise they would form a 3^--configuration containing a 2-configuration, which is impossible.

For a fixed (e_1, e_2), there are at most $(k-3)$ such edges e', since otherwise these edges together would form a k-configuration. Therefore, the number of t-sets T in (e_1, e_2) that are contained in some edge $e' \notin C$ is at most $(k-3)\binom{t}{t} = (k-3)$.

To prove assertion 2, we first note that neither e' nor f' can intersect any edge of C in more than $t-1$ vertices, otherwise e' (or f') would lie in C, which would imply $(e', f') \subseteq C$. Thus $|V(e_1, e_2) \cap V(e', f')| \le 4(t-1)$.

We now bound the number of such 2-configurations (e_1, e_2). Let R be a 3-configuration in C containing (e_1, e_2). To construct R, it suffices to find an edge $e_3 \in C \setminus \{e_1, e_2\}$ that shares at least t vertices with one of e_1 or e_2, and set $R = (e_1, e_2, e_3)$. Such an edge e_3 always exists by the definition of a t-tight component. Indeed, ordering the edges of C as $(g_1, \ldots, g_m)$ with $m \ge 3$, suppose $e_1 = g_l$ and $e_2 = g_s$ with $l < s$. If $l = 1$, then g_2 or g_3 provides a suitable choice for e_3. If $l > 1$, then by definition of t-tightness there exists some $i < l$ such that $|V(g_i) \cap V(g_l)| \ge t$, and we may take $e_3 = g_i$.

Let $\mathcal{N}(R, C)$ be the collection of 2-configurations that are not contained in C and whose vertex sets intersect $V(R)$ in at least t vertices. Let $\mathcal{M}$ be a maximal pairwise edge-disjoint subcollection of $\mathcal{N}(R, C)$. We claim that $|\mathcal{M}| \le \frac{k-5}{2}$.

If $|\mathcal{M}| > \frac{k-5}{2}$, choose $\frac{k-3}{2}$ members $S_1, \ldots, S_{(k-3)/2}$ of $\mathcal{M}$. Then the union $R \cup S_1 \cup \cdots \cup S_{(k-3)/2}$ contains $3 + 2 \cdot \frac{k-3}{2} = k$ edges, and at most $(3r - 2t) + (2r - t)\frac{k-3}{2} - \frac{k-3}{2}t = k(r-t) + t$ vertices, which forms a k-configuration, contradicting the k-free assumption. Hence $|\mathcal{M}| \le \frac{k-5}{2}$.

By maximality of $\mathcal{M}$, every 2-configuration $S \in \mathcal{N}(R, C)$ shares an edge with some member of $\mathcal{M}$. By Claim 1, the maximum number of 2-configurations containing a fixed edge is $k-2$. Thus $|\mathcal{N}(R,C)| \leq 2(k-2)\,|\mathcal{M}| \leq (k-2)(k-5)$.

Since each 2-configuration contributes at most $\binom{4t-4}{t}$ t-sets, we obtain the bound in assertion 2.

Therefore, each 2-configuration in C contributes at least

$$\binom{2q}{t} - 2\binom{q}{t} - (k-3) - (k-5)(k-2)\binom{4t-4}{t}$$

distinct t-sets to $\mathcal{B}$.

Multiplying by the total number of 2-configurations in C, we have

$$\begin{aligned}
|\mathcal{G}_3(C)| &\geq (|C|-1)\left[\binom{2q}{t} - 2\binom{q}{t} - (k-3) - (k-5)(k-2)\binom{4t-4}{t}\right] \\
&\geq (|C|-1)\frac{3}{2}\alpha && \text{by Claim 2} \\
&= |C|\alpha + \frac{1}{2}\alpha|C| - \frac{3}{2}\alpha \\
&\geq |C|\alpha && \text{as } |C| \geq 3
\end{aligned}$$

as desired.

□

3 Reference

References

[1] William G Brown, Pál Erdős, and Vera T Sós. Some extremal problems on gamma-graphs. 1973.

[2] Stefan Glock, Felix Joos, Jaehoon Kim, Marcus Kühn, Lyuben Lichev, and Oleg Pikhurko. On the (6, 4)-problem of brown, erdős, and sós. *Proceedings of the American Mathematical Society, Series B*, 11(17):173–186, 2024.

[3] Shoham Letzter and Amedeo Sgueglia. On a problem of brown, erdős and sós. *Proceedings of the American Mathematical Society*, 153(07):2729–2743, 2025.